Drona SwarmSim-Lite — Verified Evidence Report

Architecture

Rendered overhead image → ArUco/PnP → alpha-beta filter → safety →
cascaded PID → Pluto RC → PlutoInterface → PyBullet backend

Fixed parameters

N=4 • arena x=[-1,1], y=[-0.6,0.6], z=[0.6,1.4] m • min separation 0.35 m
physics 240 Hz • control 25 Hz • camera 30 FPS • 1280×720 • fx=fy=900 px
ArUco DICT_4X4_50, 0.08 m • 80 ms latency • 3 px noise • 5% dropout
drift ±0.15 m/s² per axis • 100%→93% battery scale over 120 s

Implementation

Physical UV-mapped marker planes are rendered by PyBullet. Runtime PnP uses a
finite-pose fallback for the fronto-parallel IPPE edge case. Truth is used only
for rendering and offline evaluation.

Planner, Validator and Core Mathematics

Mission: camera bootstrap → takeoff → quintic transition → 5 s hold → 3 wave cycles → 3 s hold → landing. Persistent minimum-travel slot assignment is used.

Wave: $z=1+0.2\sin(\omega t-\varphi)$, $\omega=2\pi/6$ rad/s; max $|v_z|=0.2094$ m/s; max $|a_z|=0.2193$ m/s².

N=10 validator (verified rejection)

Line: drone 0 $x=-2.025$ m outside geofence; required width ≥3.150 m or reduce N.
Circle: minimum separation 0.278 m; radius ≥0.566 m required, beyond fixed y arena.

Camera: $s_{px}=fL/(3-z)$; at $z=0.6$ m $s\approx30$ px, at $z=1.4$ m $s\approx45$ px.
Depth is more sensitive to apparent-side error than lateral position.

Hover linearization: $a_x\approx g\theta$, $a_y\approx-g\varphi$. Cascaded PID uses conditional integral anti-windup and RC bounds: tilt ±15°, yaw ±90°/s.

Vision, Safety and Fault Evidence

Camera diagnostics retain capture/delivery timestamps, detector corners, pose and acceptance status. PnP candidates with non-finite depth are rejected before filter mutation.

F1 verified line drill

Marker ID 2 hidden for 0.4 s. Logged STALE at 10.1625 s and LOST at 10.3208 s. Affected drone received controlled LAND; mission paused after LOST.

F3 verified line drill

One reported 1 m jump for marker ID 1 was rejected at 15.1208 s before filter mutation. RC remains bounded by PlutoInterface.

Safety rules: STALE→affected HOLD; LOST→affected LAND plus mission pause; separation→all HOLD_THEN_LAND; geofence→affected LAND.

Measured Results, Limitations and Reproduction

Metric	Line F1/F3 run	Circle normal run
Tracking RMS (m)	0.090	0.424
Shape RMS (m)	0.021	0.293
Min true separation (m)	0.446	0.421
Localization RMS (m)	0.086	0.086

Limitations (honest): circle tracking RMS exceeds the 0.10 m target; phase metric was not available because wave-phase fitting requires an unpaused, measured wave segment. This PDF reports evidence and does not claim those targets passed.

Reproduce: source .venv/bin/activate; PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 python -m pytest -q
python preflight_validator.py --count 10 --json
See REPRODUCIBILITY_MANIFEST.md for full commands and artifacts.